

Electrolysis

- decomposition of an ionic compound, when molten or in aqueous solution, by the passage of an electric current.

Key words (simple electrolytic cell) :

electrode the rod of metal/graphite through which electricity flows in/out of the electrolyte.

electrolyte the molten/aqueous substance that undergoes electrolysis

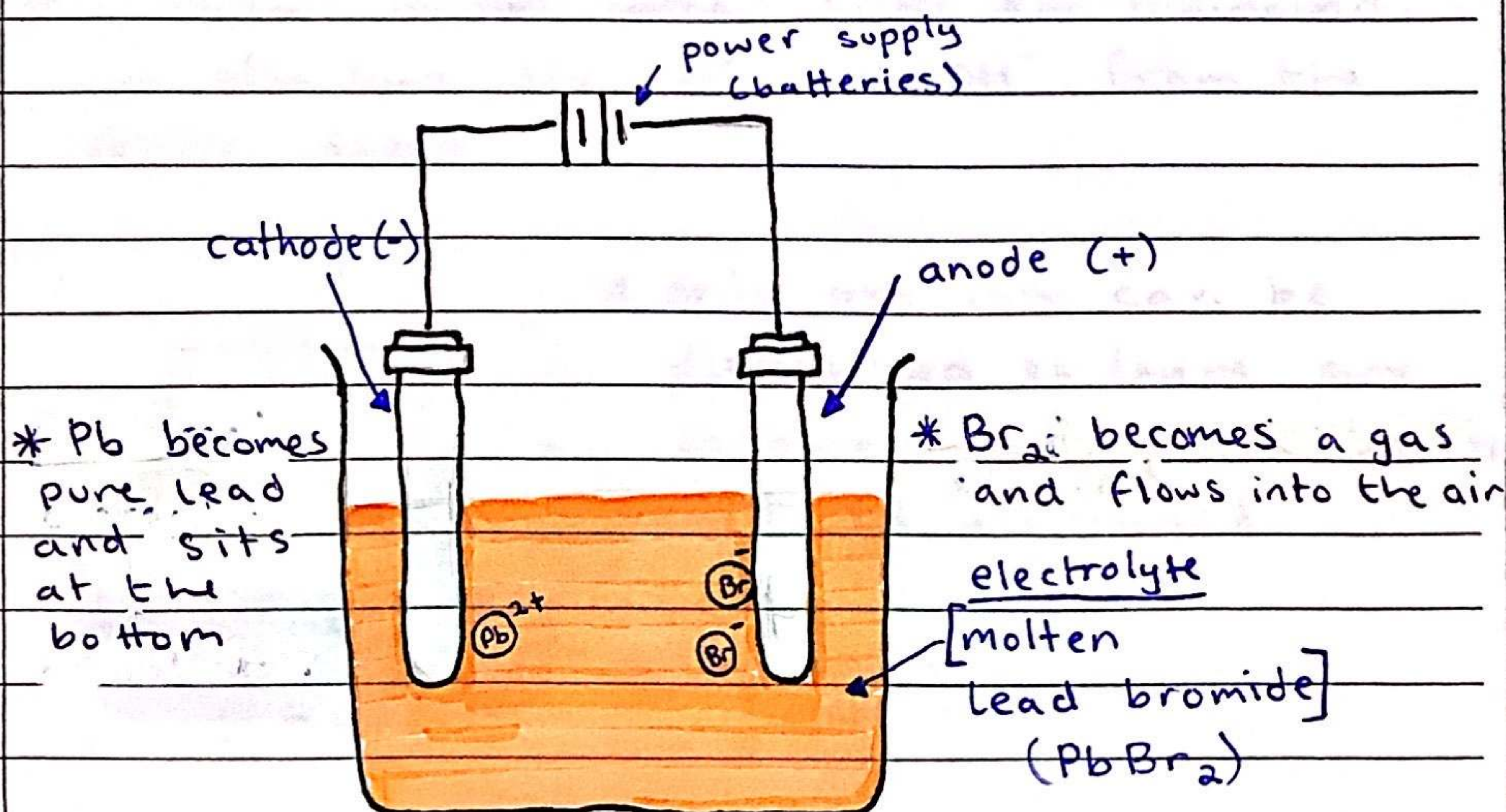
anion negatively charged ion that is attracted to the anode

cation positively charged ion that is attracted to the cathode

anode the positive electrode of an electrolysis cell

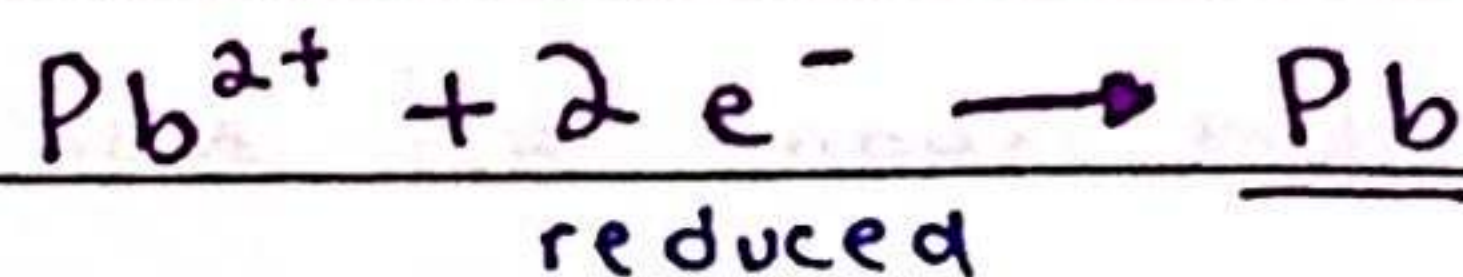
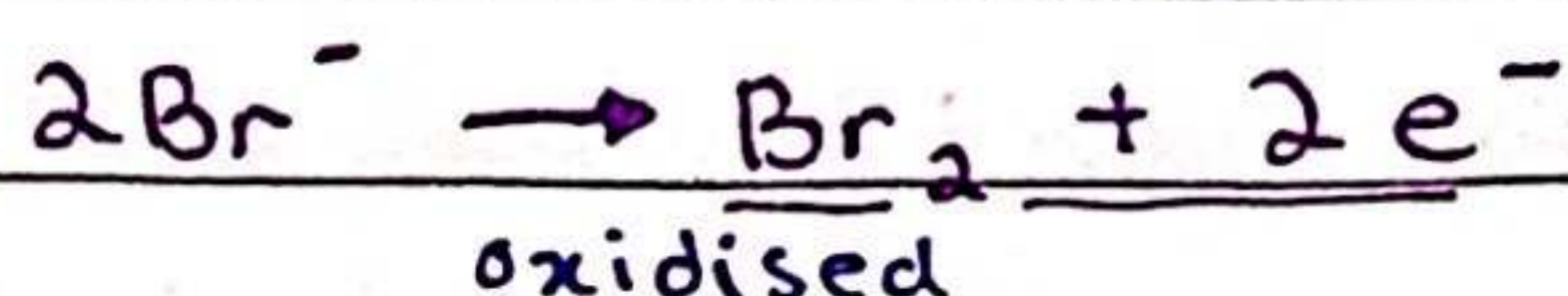
cathode the negative electrode of an electrolysis cell

* opposite charges attract each other



* the ions get discharged after reaching the anode or cathode

* "discharged" means to make a charged ion into a neutral atom



→ the electrons taken from the "Br" are taken through the current to the "Pb"

Please :: potassium

Learning :: Lead

Stop :: sodium

How :: hydrogen

Calling :: calcium

Copper :: copper

Me :: magnesium

Must :: mercury

A :: aluminum

Save :: silver

Zebra :: zinc

Gold & :: gold

Instead :: iron

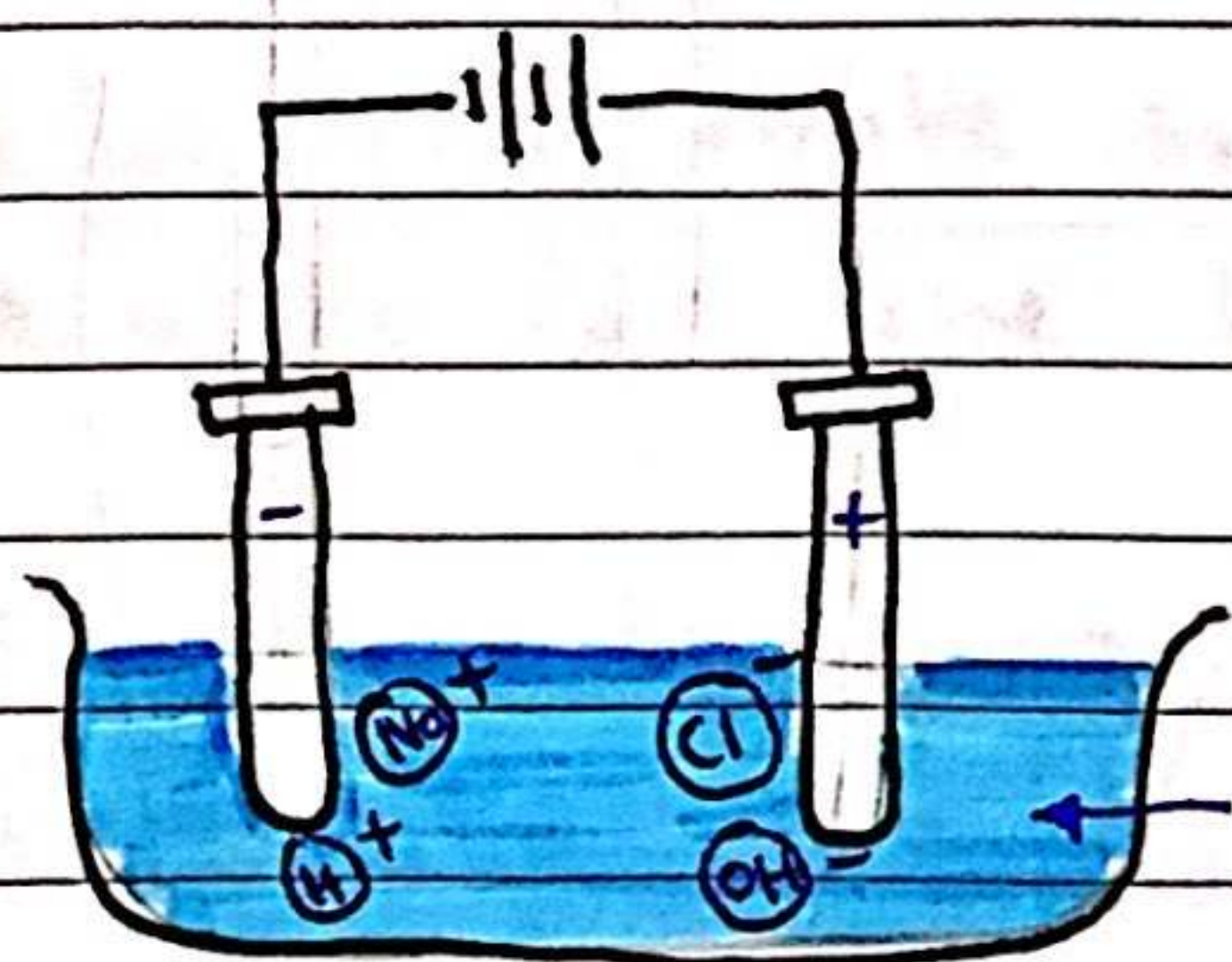
Platinum :: platinum

Try :: tin

* potassium $\xrightarrow{\text{most to least reactive}}$ platinum

→ in aqueous solution, rather than the compounds, you also have the H^+ and OH^- from the water itself

* only one ion can be discharged so there are certain rules to find which will be discharged.



aqueous sodium chloride

* to find out which positive ion the cathode will discharge:

→ the least reactive element gets discharged

* to find out which negative ion the anode will discharge:

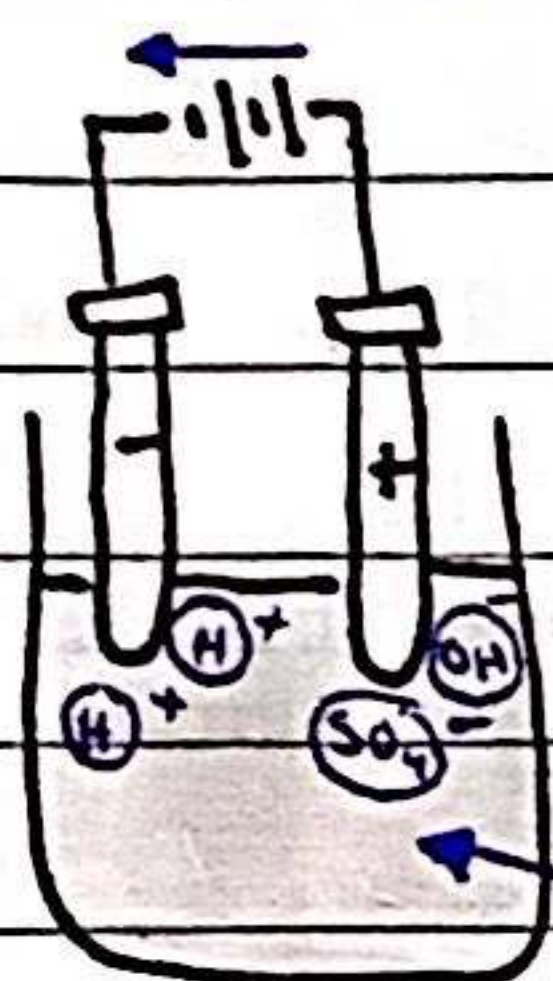
→ if a halide is present, then it will be the one to get discharged, unless it is a dilute solution e.g. Fluoride, chloride, bromide etc.

→ if a halide isn't present, then it's always the hydroxide ion that gets discharged

→ "dilute" = more water than substance

* metals & hydrogen are formed at the cathode

* non-metals are formed at the anode



ions: H^+ SO_4 + H^+ OH^-

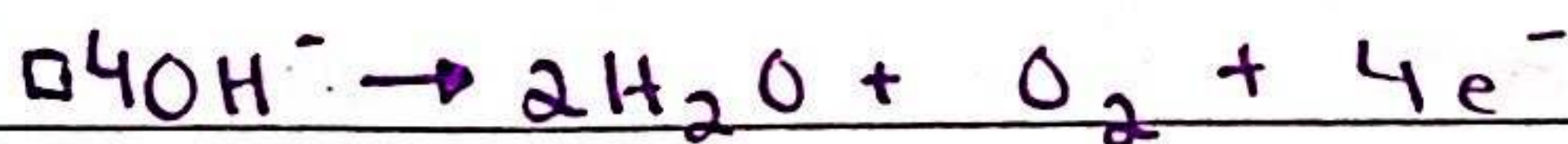
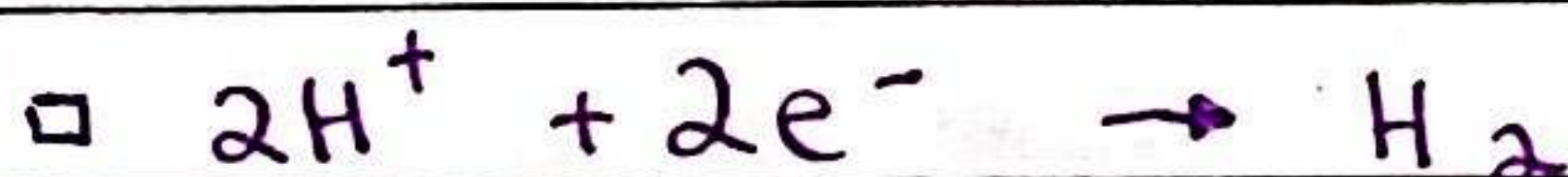
discharged: one of the 'H' ions & the 'OH' ion

Half equation

• it is the oxidation or reduction reaction

* for the dilute sulfuric acid

→ write for the ion that is discharged



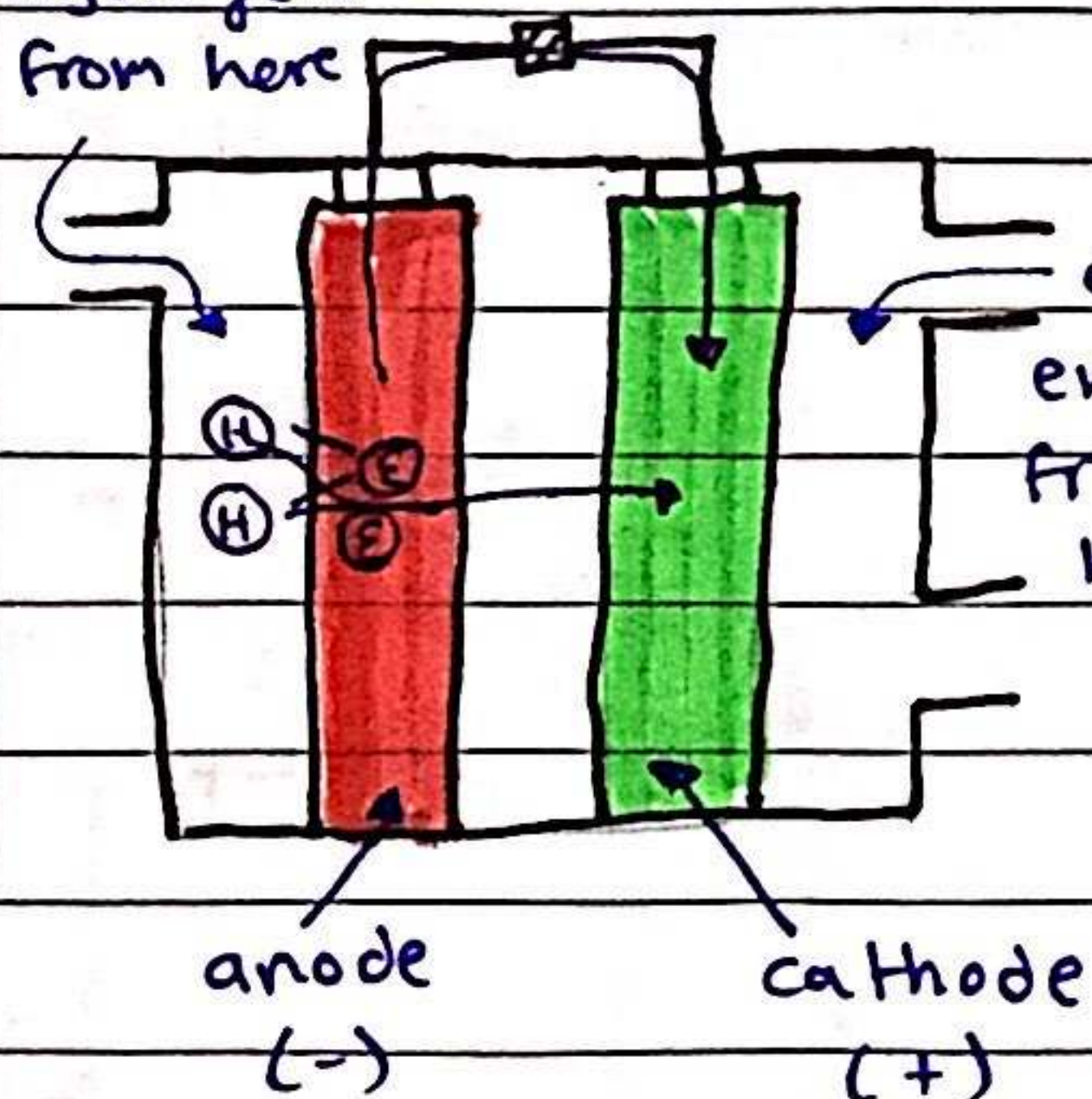
Electroplating

- the process where the surface of one metal is coated with a layer of a different metal.
- it is done to improve their appearance and resistance to corrosion
- electrolysis can be used to electroplate metals

Hydrogen - oxygen fuel cells

- fuel cells are electrochemical cells
- electrochemical cells convert energy between chemical & electrical
- more precisely, it converts the chemical energy of [fuel + oxygen] into electrical energy.
- a hydrogen-oxygen fuel cell uses hydrogen and oxygen to produce electricity with water as the only chemical product.

hydrogen enters from here



→ as fuel enters the cell, oxygen becomes oxidised which sets up a potential difference across the cell which drives the electrons through circuit. The 'H' combines with 'O' to make water (H_2O)

Pro's

- + only requires 'H' & 'O'
- + doesn't produce waste
- + lasts longer than batteries
- + less polluting to dispose of

Con's

- 'H' is a gas so takes more space to store
- explosive when mixed with air
- making 'H' fuel requires energy which often comes from fossil fuels